

Skills Summary

Limits by Factoring and Rationalizing

Use this reference while completing your worksheet or when studying.

The whole method in one line: if substitution gives $\frac{0}{0}$, the numerator and denominator share a factor that is zero at the target. Cancel it, then substitute. The cancelled function differs from the original at exactly one point — and a limit never looks at that point.

Skill 1 — Factor and Cancel a Common Factor

Patterns worth recognising: $a^2 - b^2 = (a - b)(a + b)$, $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, and $x^2 + bx + c = (x + p)(x + q)$ with $p + q = b$, $pq = c$. If $x \rightarrow k$ gives $\frac{0}{0}$, then $(x - k)$ is a factor of both.

Example: Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

Step 1. Substituting 5 gives $\frac{0}{0}$, so $(x - 5)$ divides both parts — factor and cancel rather than substitute.

Step 2. $\frac{x^2 - 25}{x - 5} = \frac{(x - 5)(x + 5)}{x - 5} = x + 5$ for $x \neq 5$, and $5 + 5 = 10$.

$$\Rightarrow \lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = 10$$

Try it: Evaluate $\lim_{x \rightarrow 6} \frac{x^2 - 36}{x - 6}$

check: 12

Skill 2 — Rationalize with the Conjugate

Rule: when a square root blocks the cancelling, multiply top and bottom by the conjugate of the part that contains it, using $(\sqrt{A} - B)(\sqrt{A} + B) = A - B^2$. The radical disappears from that part and the blocking factor appears.

Example: Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$

Step 1. Substituting 0 gives $\frac{0}{0}$ and nothing factors, so clear the radical with the conjugate $\sqrt{x+4} + 2$.

Step 2. $\frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2}$, and at $x = 0$ that is $\frac{1}{4}$.

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} = \frac{1}{4}$$

Try it: Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+25} - 5}{x}$

check: $\frac{1}{10}$

(!) **Watch out:** $\sqrt{x+4} \neq \sqrt{x} + 2$. Splitting a root over a sum is the single most common way this method goes wrong. Multiply by the conjugate instead — that is the only legal move.

Skill 3 — Reading a Table Before the Algebra

What a table can and cannot do: inputs approaching the target from both sides give a strong prediction of the limit, and they catch a sign or arithmetic slip fast. They are evidence only — a finite list of samples cannot prove the value. Algebra proves it; the table tells you what to expect.

Example: A table for $h(x) = \frac{x^2-1}{x-1}$ reads 1.99 at $x = 0.99$ and 2.01 at $x = 1.01$. Find $\lim_{x \rightarrow 1} h(x)$ exactly.

Step 1. The two-sided table points at 2, so I expect 2 — now prove it, because a table is never the proof.

Step 2. $\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1$ for $x \neq 1$, and $1+1 = 2$, matching the prediction.
 $\Rightarrow \lim_{x \rightarrow 1} h(x) = \mathbf{2}$

Try it: Evaluate $\lim_{x \rightarrow 1} \frac{x^3-1}{x-1}$ exactly, then predict what a table near $x = 1$ would show.

check: